

Biology, 2018 EC, entrance exam, round 1

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 94

Total marks: 94

Name _____ Grade _____ ID No. _____ Date _____

1. A city Municipality following the ban of non-degradable plastic bags, proposed to use genetically engineered bacterium to degrade plastic wastes for making bags. However, an authority claims the use of bioengineered bacteria due to the fear of its negative consequences on the environment. Based on this scenario, which decision is valid? (1)
 - A. accepted with stringent protocols to reduce its effect.
 - B. rejected since the city can use already existing materials for packing.
 - C. accepted whatever the environmental consequences it caused.
 - D. rejected since the municipality can use the native species.
2. This item is based on the following mechanism of hearing in human ear. I. Nerve impulses produced by hair-like sensory cells are transmitted to, and interpreted by the brain. II. Ear ossicles amplify vibrations and cause vibration of the ear fluid. III. Sound wave from the air is collected by pinna and directs to the ear canal. IV. Vibrations are magnified in eardrum and transmitted to oval window. Then which of the following order summarizes correct path of hearing in human ear? (1)
 - A. III, I, II, IV
 - B. III, II, IV, I
 - C. III, IV, II, I
 - D. III, IV, I, II
3. Which one of the following is the industrial importance of insects? (1)
 - A. Serving as pollinating agents for plants
 - B. Producing silk fibers for textiles
 - C. Controlling pests that attack crops
 - D. Serving as the source of pain-relieving drugs
4. Two processes were taking place inside a eukaryotic cell. In process A, a ribosome made a specific protein by reading the instructions from mRNA. In process B, an mRNA was synthesized inside the nucleus. Based on this scenario, which of the following processes correctly 2018 Biology First Round represents A and B respectively? (1)
 - A. Translation and Replication
 - B. Transcription and Translation
 - C. Replication and Translation
 - D. Translation and Transcription
5. Which one of the following statements is correct about regulation of enzyme through covalent modification? It happens through the (1)
 - A. permanent binding of effectors to the active site of an enzyme.
 - B. addition or elimination of some molecules attaching of enzyme protein.
 - C. reversible binding of effectors to the allosteric site of an enzyme.
 - D. inhibition of the enzyme by the end product of a pathway.

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6. A farmer crossed a heterozygous insect-resistant maize with a non-insect resistant maize variety, where insect-resistance gene is dominant over non-resistance one. What is the probability of the offspring resulted from this cross? (1)
- A. 75% resistant and 25% are non-resistant to insect infestation.
 - B. 50% resistant and 50% are non-resistant to insect infestation.
 - C. 25% resistant and 75% are non-resistant to insect infestation.
 - D. 100% of the offspring are resistant to insect infestation.
-
7. Which statement does NOT show the role of biological discoveries for the society? (1)
- A. Detection of cancer cells through advanced instruments
 - B. Collecting and analyzing evidences from crime scene
 - C. Studying of microorganisms that cause disease to humans
 - D. Resolving conflict that happens as a result of over population
-
8. The function of ovaries in female reproductive system differs from testes in the male reproductive system in which ovaries produce 2018 Biology First Round (1)
- A. estrogen that initiates pregnancy, whereas testes produce LH and sperm.
 - B. progesterone that initiates ovulation, whereas testes produce FSH and sperm.
 - C. estrogen that promotes the maturation of the ovum, whereas testes produce testosterone and sperm.
 - D. androgen that develops a mature ovum, whereas testes produce estrogen and sperm.
-
9. A following biology guide prepared a thin slice of onion epidermal cells and placed it in a hypertonic solution for 15 minutes. What would be the result of the experiment? They (1)
- A. become crenated due to damage to the cell walls by the water pressure.
 - B. burst as the vacuoles presses against the cytoplasm and cell membrane.
 - C. get plasmolyzed due to excessive loss of water from the vacuole.
 - D. become turgid due to net inflow of water from the external solution.
-
10. Which of the following is correct regarding photorespiration? It is a process of (1)
- A. balancing O_2 and CO_2 concentration in the cell.
 - B. oxidizing organic compounds to release CO_2 .
 - C. respiration common in plants during the daytime.
 - D. reduction of organic compounds to generate ATP.
-
11. Which alternative contains the behavioral method of thermoregulation with its example? (1)
- A. Honey bees - stay in hive to survive cold seasons
 - B. Cats - use involuntary muscle contraction to generate heat
 - C. Foxes - narrow blood vessels to prevent heat loss
 - D. Rodents - sweat to maintain body temperature
-
12. Why hillside terracing is considered as a mechanical soil conservation measure? Because it (1)
- A. is constructed to intercept runoff water that drain from the slopy area.
 - B. breaks largely the speed of wind to avoid wind erosion.
 - C. is a reservoir of rainwater to use the water for irrigation purpose.
 - D. is constructed to divert the rainwater to other area with higher slope.

13. The amount of energy required to raise the temperature of one gram of water by one degree Celsius is (1)
- A. cohesion force. B. surface tension.
C. latent heat of vaporization. D. specific heat capacity.

14. This item is based on the following table about the classification of organic molecules. No. Organic molecules Elemental composition Monomers 1. Sucrose Carbon, hydrogen and oxygen Glucose and fructose 2. Starch Carbon, hydrogen and oxygen Glucose and galactose 3. Proteins Carbon, hydrogen and oxygen Amino acids 2018 Biology First Round No. Organic molecules Elemental composition Monomers 4. Lipids Carbon, hydrogen and oxygen Fatty acid and glycerol Which numbers in the above table correspond to the correct elemental composition and monomers of the organic molecules? (1)

[Figure: A table with four columns titled No., Organic molecules, Elemental composition, and Monomers. Row 1 lists Sucrose with Carbon, hydrogen and oxygen, and Glucose and fructose. Row 2 lists Starch with Carbon, hydrogen and oxygen, and Glucose and galactose. Row 3 lists Proteins with Carbon, hydrogen and oxygen, and Amino acids.]

- A. 1, 2 and 4 B. 1, 3 and 4
C. 1 and 3 D. 1 and 4

15. What are ethical issues in biology? They are issues linked to (1)
- A. use of microscope to observe human cheek cells.
B. study the interaction between organisms.
C. use of scientific discoveries for human wellbeing.
D. study the role of evolution in speciation.

16. Which of the following is a symptom of gonorrhea infection? (1)
- A. Painful urination and abnormal discharge from the penis
B. Necrotizing genital ulcer accompanied by lymphadenopathy
C. Sores around the anus that leads to cancer through time
D. Rash on hands and soles of feet with late neurological complication

17. All the synthesis and breakdown chemical reactions together occurred in our cells is (1)
- A. catabolism. B. anabolism.
C. respiration. D. metabolism.

18. Based on the following diagram of the human heart, identify the flow and anatomical parts accordingly. Addis Aptitude አጥን በመጠቀም ለSAT ፈተና ዝግጅት ያድርጉ። ለጎኑን በመጠቀም የቴሌቢዎን ቻናላችንን ይቀላቀሉ። @addis_apptitude 2018 Biology First Round (1)

[Figure: A cross-sectional diagram of the human heart showing internal chambers, valves, and major blood vessels with directional arrows indicating blood flow. The right side of the heart (colored blue) and left side (colored red) are labeled with numbers. Number 1 points to the right atrium, which receives blood from superior and inferior vena cavae (number 11 pointing to inferior vena cava, and number 10 pointing to the right ventricle). Number 6 points to the pulmonary veins and left atrium on the upper right side. Number 9 points to the left ventricle. A large aortic arch in the center has an upward-pointing arrow. Blue pulmonary arteries branch to the lungs with outward arrows. Inside the heart, white arrows show blood flowing through the tricuspid and bicuspid (mitral) valves into the ventricles, and out through the pulmonary trunk and aorta.]

- A. Left atrium receives deoxygenated blood
B. Right ventricle pumps oxygenated blood
C. Aorta distributes oxygenated blood to the body
D. Pulmonary vein carries deoxygenated blood

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19. A biologist observed the sister chromatids that lined up end-to-end at the equator of the cell in a sample taken from Test tube 1, whereas in another cell sample (Test tube 2) she observed the sister chromatids are pulled to opposite poles of the equator. Based on this information, which stages of meiosis are observed in test tube 1 and 2? (1)
- A. Prophase I and Metaphase I
B. Metaphase II and Anaphase II
C. Anaphase II and Telophase II
D. Prophase I and Telophase I
-
20. Suppose the fibers in the connective tissue of the skin and blood vessels do not form properly and causes bleeding gums and poor healing of wounds. This problem is related with deficiency of (1)
- A. Vitamin D.
B. Vitamin A.
C. Vitamin B.
D. Vitamin C.
-
21. What would happen to a plant placed in areas with unilateral light? (1)
- A. The light exposed side would produce more root hairs.
B. The light exposed parts would produce more plant hormones. 2018 Biology First Round
C. The plant would bend towards the light source.
D. The plant would immediately stop growth.
-
22. In a grassland, the gazelle population initially increase in fast pace, then the growth rate reduces gradually and finally levels-off when the population reaches its maximum. This population growth model shows (1)
- A. exponential because the gazelle experiences maximum per capita rate of increase.
B. logistic because the size of gazelle population reaches the carrying capacity.
C. logistic because the per capita rate increase decreases as the population decreases.
D. exponential since the growth rate of the gazelle reach maximum at the carrying capacity.
-
23. Grade 9 students identified an organism using dichotomous keys. The organism was poikilothermic, invertebrate and had six jointed legs. To which phylum would the students group this organism? (1)
- A. Arthropoda
B. Mollusca
C. Porifera
D. Cnidaria
-
24. Which one of the following statements is correct about innate immunity? It (1)
- A. produces antibody that neutralizes antigens in the blood.
B. responds better after the second encounter of a pathogens.
C. has memory to recognize pathogens.
D. is a non-specific host defense system.
-
25. Which of the following is NOT a pollinating agent? (1)
- A. Birds
B. Insects
C. Wind
D. Microbes
-
26. Which of the following is the major causes of evolution that results from a sudden, and random change of DNA sequence? (1)
- A. Artificial selection
B. Induced mutation
C. Environmental stability
D. Acquired characteristic

27. Gardener brings an ornamental plant from other country and plant it in urban parks. This species through time moved into conservation area and caused a loss for many other plant species. This is a loss of biodiversity due to (1)
- A. invasive species. B. climate change.
C. deforestation. D. overexploitation.
28. Which one of the following breathing structures and its function is correctly matched? (1)
- A. Trachea - prevents the collapse of respiratory structures
B. Bronchioles - deliver air from the nasal cavity to the bronchi
C. Bronchi - exchange gases between capillaries in the lungs
D. Larynx - prevents choking by covering the opening in the trachea 2018 Biology First Round
29. Suppose a person's height and body mass are 1.81 m and 68 kg, respectively. Which one of the following categories best describes the person? She is (1)
- A. overweight. B. normal.
C. obese. D. underweight.
30. Which one of the following statements is correct about complete metamorphosis? The (1)
- A. larvae develop directly into adult B. larvae develop into nymph
C. eggs hatch into larvae after few days D. nymph develops into adult
31. Plant cells differ from animal cells in that, they have (1)
- A. larger ribosomes for protein synthesis. B. large central vacuoles in their cells.
C. cell membrane with double layers. D. a pair of centrioles for cell division.
32. This item is based on the following table about food chain. No. Characteristics Trophic level Feeding level (1)
- Example 1. Make their own food First Producers Grass 2. Feeds on herbivores Third Tertiary consumer Frog 3. Feeds on another carnivore Fourth Tertiary consumer Snake 4. Feeds on green plants Second Primary consumer Grasshopper
- Which numbers in the above table contain the correct description of a food chain with its examples?
- [Figure: A table with four columns titled No., Characteristics, Trophic level, Feeding level, and Example. Row 1 lists No. 1, Characteristics: Make their own food, Trophic level: First, Feeding level: Producers, Example: Grass. Row 2 lists No. 2, Characteristics: Feeds on herbivores, Trophic level: Third, Feeding level: Tertiary consumer, Example: Frog. Row 3 lists No. 3, Characteristics: Feeds on another carnivore, Trophic level: Fourth, Feeding level: Tertiary consumer, Example: Snake. Row 4 lists No. 4, Characteristics: Feeds on green plants, Trophic level: Second, Feeding level: Primary consumer, Example: Grasshopper.]
- A. 1, 2 and 3 B. 1 and 2
C. 2 and 4 D. 1, 3 and 4
33. Which one of the following bacteria are correctly grouped into their category based on their shape? (1)
- A. Bacilli - rod shaped bacteria B. Cholera - spiral shaped bacteria
C. Spirochaetes - spherical bacteria D. Cocci - corkscrew shaped bacteria
34. Which one of the following nutrients is correctly paired with its deficiency diseases? (1)
- A. Phosphorous - kwashiorkor
B. Protein - marasmus 2018 Biology First Round
C. Calcium - rickets
D. Iron - bleeding gums

-
- 35.** In the presence of oxygen, the cell theoretically yields approximately 36-38 ATP per glucose. Justify why the majority of this energy is harvested by electron transport system (ETS) rather than by glycolysis or the Krebs cycle? Because ETS (1)
- A. is the sole stage where glucose is directly converted into ATP through substrate level phosphorylation.
 - B. uses the stored potential energy of NADH and FADH₂ to create proton gradient that drives ATP synthase.
 - C. occurs in the cytosol where there is more space to produce ATP abundantly.
 - D. utilizes oxygen that plugs the leak in the membrane, leading to produce more metabolic energy.
-
- 36.** Which of the following is correct about gene therapy? It is a genetic technique that helps to (1)
- A. replace the whole genome of a person with genetic disorder.
 - B. enhance transfer of genetic disorders from parents to their offsprings.
 - C. identify disorders that passes from parents to offsprings.
 - D. treat genetic disorders by transferring sections of DNA to patient's cells.
-
- 37.** Which structure and function of the types of teeth is correctly matched? (1)
- A. Molars - broader teeth specialized for grinding foods
 - B. Premolars - flat teeth specialized for tearing foods
 - C. Incisors - flattened teeth specialized for crushing foods
 - D. Canine - sharp teeth specialized for cutting foods
-
- 38.** Among the following features, which one describes kingdom fungi? They (1)
- A. are multicellular and have chemoautotrophic mode of nutrition.
 - B. ingest and digest their food by enzymes in their body.
 - C. undergo extracellular digestion to obtain their nutrition.
 - D. are prokaryotic and cosmopolitan in their distribution.
-
- 39.** Which of the following is the correct definition of mutation? It is a (1)
- A. spontaneous variation in the spatial distribution of organisms.
 - B. random change in genetic information of organisms.
 - C. gradual change in the phenotype of the organisms.
 - D. gradual variation in the spatial distribution of organisms.
-
- 40.** Suppose a biology teacher take a sample of bacteria from a food and dissolve in a distilled water. If she wants to isolate the bacterium, what is the next direct procedure to accomplish her task? (1)
- A. Conduct gram staining to identify their feature
 - B. Count a colony and harvest target product
 - C. Perform serial dilution and grow in media
 - D. Identify bacterial morphology and grow in media
-
- 41.** In which of the following options, a hormone is correctly paired with its function? (1)
- A. Norepinephrine - facilitate the secretion of glucagon
 - B. Oxytocin - helps during delivery of the baby
 - C. Parathyroid - control iodine levels in blood stream
 - D. Aldosterone - increase heart rate and force of contraction
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42. This item is based on the following events of maintaining water balance in the body. I. Second coiled tubules of kidney become permeable to reabsorb more water from the kidney II. High ADH is released into the blood III. Produce large volume of diluted urine due to less reabsorption of water by kidney IV. Very few ADH released to the blood Based on the above list, which of the following numbers shows the mechanism of water balance in our body correctly? Numbers
- A. I and IV are for water conserving, II and III are for water removing.
B. III and IV are for water conservation, I and II are for water removing.
C. I and IV are for water removing, II and III are for water conserving.
D. III and IV are for water removing, I and II are for conserving water.
-
43. What is the trophic level? It is the
- A. complex interaction between organisms in a food web.
B. interaction of organisms with the abiotic components.
C. linear feeding relationship between organisms.
D. specific position an organism occupies in a food chain.
-
44. A grade 12 teacher demonstrated how light travelled through the eye to form an image on the retina using an eye model. Which one of the following results shows the correct sequence of this process?
- A. Reflected light → pupil → lens → optic nerve → retina → auditory nerve → brain
B. Reflected light → sclera → choroid → lens → retina → optic nerve → brain
C. Reflected light → cornea → pupil → lens → retina → optic nerve → brain
D. Reflected light → cornea → pupil → lens → retina → olfactory nerve → brain
-
45. What is the main function of the cell membrane? It controls
- A. transport of molecules.
B. reproduction of cells.
C. synthesis of lipids.
D. protein synthesis.
-
46. A scientist isolated three different protein molecules (X, Y and Z) from a medicinal plant as follows. X molecule - is a segment of a protein folded into alpha-helix and beta-pleated sheet. Y molecule - is an enzyme with its protein folded into three-dimensional structure. Z molecule - is a linear protein with specific amino acid sequence. Which molecule contains primary, secondary and tertiary structures of protein, respectively?
- A. Y, Z and X
B. Y, X and Z
C. X, Z and Y
D. Z, X and Y
-
47. A student, following appropriate laboratory procedures, testing for the presence of starch in green plant leaves. He immersed the leaf in hot water for 30 seconds in one of the steps just before adding iodine. Why did he immerse the leaf into the hot water? To
- A. remove the debris and facilitate gas exchange
B. decolorize the leaf and make it easy for the test
C. denature the enzyme and make the leaf permeable
D. improve the visibility of various pigment colors
-
48. Which one of the following statements explains the effects of drinking too much alcohol? It
- A. exposes to constipation and esophageal disease.
B. makes hyperactive and constricts the blood vessels in the heart.
C. enables to make the right decision without fear.
D. exposes to convict domestic violence incidents.

49. A survey in rural areas of Ethiopia discovered several people with enlarged body parts especially the limbs. Which one of the following is correct about the disease? It (1)
- A. happens due to a genetic disorder that causes accumulation of fat in the lymphatic system.
 - B. is caused due to bone marrow cancer that leads to deposition of fluids in the lymphatic system.
 - C. happens due to deficiency of vitamins that protects lymphatic systems from damage.
 - D. is caused by parasitic worms that blocks the lymphatic system leading to accumulation of fluids.
-
50. Why do you sense it in microsecond when a tip of your toe is knocked by obstacles? Because the information is transmitted (1)
- A. through electrochemical signal traveling along a neuron.
 - B. through neural axon when it is in resting potential phase.
 - C. by hyperpolarizing the neuronal membrane potential.
 - D. due to equal charge distribution across the membrane of the axon.
-
51. This item is based on the following list of information about route of transport of water in plants. I. Entry of water from stem xylem into leaf II. Entry of water into the root xylem III. Flow of water through xylem tissues in stems IV. Entry of water from soil into root hairs V. Evaporation of water from leaf surfaces into the environment Which one of the following alternatives shows the correct route of water in plants? (1)
- A. $IV \Rightarrow II \Rightarrow III \Rightarrow I \Rightarrow V$
 - B. $V \Rightarrow III \Rightarrow II \Rightarrow I \Rightarrow IV$
 - C. $V \Rightarrow I \Rightarrow III \Rightarrow II \Rightarrow IV$
 - D. $IV \Rightarrow III \Rightarrow II \Rightarrow I \Rightarrow V$
-
52. Which one of the following statements is correct about absorption of different end products of digestion? (1)
- A. The last organ that absorbs nutrients is the small intestine.
 - B. Amino acids are absorbed into the capillary networks.
 - C. The large intestine the place for most absorption to take place.
 - D. Diffusion is the only mechanism of absorbing nutrients.
-
53. Which of the following statements is correct about the Kyoto Protocol on climate change? It was an agreement that industrialized nations (1)
- A. forced developing countries to restrict their carbon emissions.
 - B. were obliged to reduce greenhouse gas emission significantly.
 - C. encouraged to combat global epidemics by natural products.
 - D. promoted urbanization in the third world countries.
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54. Concerning the interactions between solar radiation and atmospheric gases, why does the accumulation of CO₂ by the fossil fuels affect the energy after reaching the Earth, rather than before? Because (1)
- A. CO₂ is transparent to short-wave radiation but it absorbs long infrared radiation that reflects back from the surface.
 - B. light energy is converted into chemical energy upon striking the soil, and then rises and creates thermal heat.
 - C. fossil fuel that emits CO₂ is highly reduced comparing to the natural process that restores the balance.
 - D. CO₂ acts as a catalyst that makes the sunlight 10% stronger in the upper atmosphere before getting the ground.
-
55. This item is based on the following information about the process of mitosis. I. Nuclear membrane reappears II. Chromosomes line up across the equatorial plane III. Sister chromatids pull apart by mitotic spindle IV. Chromosomes become thick and visible Which one of the following alternatives contains the correct order of the above mitosis events? (1)
- A. $IV \rightarrow II \rightarrow III \rightarrow I$
 - B. $I \rightarrow II \rightarrow IV \rightarrow III$
 - C. $I \rightarrow IV \rightarrow III \rightarrow II$
 - D. $IV \rightarrow III \rightarrow II \rightarrow I$

56. This item is based on the following information about urine formation. I. Reabsorption of small solutes at the proximal convoluted tubule. II. Selective reabsorption of ions at the distal convoluted tubule. III. Filtration of the blood at the glomerulus. IV. Reabsorption of solutes and water at the collecting duct. V. Reabsorption at the loop of Henle. Which alternative contains the correct route of urine formation in the nephrons? (1)
- A. III → II → IV → V → I
B. III → I → V → II → IV
C. III → V → I → II → IV
D. III → II → I → IV → V

57. Which of the following genetically engineered crops is herbicide-resistant? A modified crop with (1)
- A. a foreign gene that passes to non-target crop species.
B. an enzyme that digest plant roots of weeds.
C. an enzyme to break down the glyphosate.
D. an inserted gene that synthesizes glyphosate in crops.

58. What is the effect of the deficiency of iodine? (1)
- A. Hepatic cirrhosis and brain damage
B. Hemosiderosis and hemochromatosis
C. Enlargement of thyroid glands
D. Elevation of blood pressure

59. Which alternative contains the correct comparison between the components in atmospheric air and exhaled air? (1)
- Atmospheric air contains
- A. 4% carbon dioxide, but exhaled air contains 0.04% carbon dioxide.
B. 21% oxygen, but exhaled air contains 16% oxygen.
C. 78% nitrogen, but exhaled air contains 68% nitrogen.
D. 15% moisture, but exhaled air contains 21% moisture.

60. Suppose a sudden earthquake starts shaking the earth while you are on a way towards your school. Which safety precaution measures should you practice? (1)
- A. Moving towards large buildings
B. Continue walking towards the school
C. Standing in open ground until it stops
D. Running back towards home

61. This item is based on the following table that shows homeotherms and poikilotherms with their respective examples. No. Features Examples 1. Have internal control over their body temperature Frogs 2. Use burrows to keep their body warm Snakes 3. Use vasoconstriction to prevent heat loss Crocodiles 4. Generate internal heat to adjust their body temperature Human Which alternative contains the numbers with the correct distinguishing features of poikilotherms and homeotherms with their corresponding examples, respectively? (1)
- [Figure: A table with three columns labeled No., Features, and Examples. Row 1 lists number 1, feature 'Have internal control over their body temperature', and example 'Frogs'. Row 2 lists number 2, feature 'Use burrows to keep their body warm', and example 'Snakes'. Row 3 lists number 3, feature 'Use vasoconstriction to prevent heat loss', and example 'Crocodiles'. Row 4 lists number 4, feature 'Generate internal heat to adjust their body temperature', and example 'Human'.]
- A. 3 and 4
B. 2 and 4
C. 2 and 3
D. 1 and 2

62. Viruses start replication and cause disease when they enter into their appropriate host. This is because they (1)
- A. lack structures to generate energy and synthesize molecules.
B. are tiny so that they lack enough space for replication.
C. are instructed by the genetic material of living organism.
D. lack genetic materials that instruct their vital activities.
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63. Student X and Y were discussing on the origin of life on earth. Student X explained the process on how rotting meat turned into flies through time. But student Y didn't accept this view since he believed that life is the inherent property of the universe that has always existed. Based on their discussion, which views of evolution was described by the students? (1)

- A. Panspermia by student Y and big bang theory by student X
- B. Biochemical view by student X and cosmozoic theory by student Y
- C. Special creationism by student Y and abiogenesis by student X
- D. Spontaneous generation by student X and eternity of life by student Y

64. During emergencies like car accident, individuals lose significant amount of blood that could potentially harm their health and even cost their life. What should be considered to donate blood in this condition? (1)

- A. The donor's blood should contain antibody that protects the recipient from pathogens.
- B. The age of the donor should be under 18 years in order to donate functional bloods.
- C. The donor's and recipient's blood groups should be compatible and safe.
- D. The recipient should be young in order to receive blood from the donor.

65. Which of the following combinations, the factor and its effect on enzyme activity is correctly paired? (1)

- A. pH - decrease kinetic energy of reacting molecules
- B. Competitive inhibitor - binds to allosteric site of an enzyme
- C. Temperature - alter the charge of amino acids
- D. Radiation - creates disorders in macromolecules

66. With regard to the contribution of soil microorganisms for soil fertility, which one of the following processes decrease nitrogen from the soil? (1)

- A. Conversion of nitrates in the soil into nitrogen gas
- B. Fixation of atmospheric nitrogen into ammonia
- C. Decomposition of organic nitrogen sources to nitrate
- D. Transformation of ammonium ions into nitrate

67. How does gene flow become a source of genetic variations? (1)

- A. Mass emigration of individuals stabilizes gene flow between source and receiving populations.
- B. Emigration of individuals increases genetic heterogeneity of a source population.
- C. Immigrants with new trait decreases allele frequencies of a receiving population.
- D. Immigrants with new trait increases genetic variability of a receiving population.

68. Consider the following diagram which indicates the parts of the human excretory system. Identify the part labeled 'M'. Addis Aptitude አገልግሎት ለSAT ፈተና ዝግጅት ያድርጉ። ሊንኩን በመጠቀም የቴሌግራም ቻናላችንን ይቀላቀሉ። (1)

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[Figure: An anatomical diagram of the human excretory system overlaid on a male torso. The diagram shows two reddish-brown bean-shaped kidneys near the upper abdomen, with major blood vessels running vertically between them (a red aorta and a blue vena cava). From each kidney, a thin tube extends downwards into a sac-like urinary bladder situated in the lower pelvic region, from which a single exit tube leads further down. A large uppercase letter M is placed on the left side of the torso, with a pointer line extending from it to indicate one of the thin tubes connecting the right kidney to the urinary bladder.]

- A. Ureter
- B. Right Kidney
- C. Urinary Bladder
- D. Urethra

69. Which one of the following is a locally available plant used for treatment of tapeworm? (1)

- A. Cucurbita pepo (Dubba) B. Lepidium sativum (Feto)
- C. Eucalyptus globulus (Nech-Bahir zaf) D. Trigonella foenumgraecum (Abish)

70. Which one of the following shows a population pyramid of age in third world countries? A pyramid with (1)

- A. broad base and narrowing from younger to older ages.
B. flattened at the middle age and wider at older ages.
C. uniform shape throughout all age groups.
D. narrower base with younger ages.

71. Recently married woman, who wants to stay for few years without having a child, consulted a gynecologist to recommend her a contraceptive with no side effects and must be physically efficient to prevent the entry of sperm to uterus. Based on her preferences, he recommends

- A. spermicides which kill sperm and prevent the sperm from meeting the egg.
B. oral contraceptive act by avoiding contact between sperm and ova.
C. condom which forms barrier and works without changing the reproductive physiology.
D. tubal ligation which prevents the contact of egg with the sperm.

72. Which of the following is correct about the cell cycle? (1)

- A. G1 phase is the stage at which cells are metabolically inactive. 2018 Biology First Round
- B. M phase is the stage at which nuclear division takes place
- C. S phase is the stage at which cytoplasmic division takes place.
- D. G2 phase is the stage at which chromosome duplicate.

73. During photosynthesis, oxygen is produced as a by-product in (1)

- A. the thylakoid through reduction of NADPH.
B. Calvin cycle by fixation of CO₂.
C. photosystem II by reduction of water.
D. light-dependent reaction by photolysis.

74. This item is based on the following table that shows the structure of plants and their corresponding functions.

No.	Plant structures	Functions
1.	Epidermis	Reduces water loss through transpiration
2.	Endodermis	Produces food by photosynthesis
3.	Vascular bundles	Conduct nutrients to different part of the plant
4.	Cortex	Prevent tissues from infection by pathogens

Which alternative contains the numbers with the correct plants structure and its function?

(1)

[Figure: The table for Question 79 has three column headers: No., Plant structures, and Functions. Row 1 lists: number 1, Epidermis, Reduces water loss through transpiration. Row 2 lists: number 2, Endodermis, Produces food by photosynthesis. Row 3 lists: number 3, Vascular bundles, Conduct nutrients to different part of the plant. Row 4 lists: number 4, Cortex, Prevent tissues from infection by pathogens.]

- A.** 1 and 3 **B.** 2 and 3
C. 1 and 4 **D.** 2 and 4

75. What is the importance of monosaccharides? They (1)

- A. mainly used as structural component of cells.
- B. serve as the precursor of vitamin D.
- C. used as stored form of energy.
- D. serves as energy source for the cell.

-
- 76.** Which one of the following behavioral patterns represent territoriality? (1)
- A. Marking the directions to food sources using pheromones
 - B. Attracting suitable mates through visual displays
 - C. Surviving in an ecosystem through division of labor
 - D. Pounding on the chest and thumping the ground with hands
-
- 77.** Which of the following is correct about pyruvate oxidation in the cell? The process involves (1)
- A. utilizing oxygen as electron acceptor during hydroxyethyl group oxidation to form acetyl CoA.
 - B. decarboxylation, oxidization of hydroxyethyl group, and formation of acetyl CoA.
 - C. carboxylation, reduction of hydroxyethyl group and formation of acetyl CoA.
 - D. oxidization of hydroxyethyl group to form acetyl CoA by utilizing oxygen.
-
- 78.** Which of the following is correct about bioremediation? It is the process of (1)
- A. detoxification and degradation of pollutants using living organisms. 2018 Biology First Round
 - B. composting using aerobic microorganisms to convert organic waste.
 - C. toxic pollutant accumulation in tissues of organisms along food chain.
 - D. producing biogas from inorganic wastes by aerobic microorganisms.
-
- 79.** Which one of the following enzymes is responsible to catalyze the joining of two molecules by hydrolysis of ATP? (1)
- | | |
|------------|-----------------|
| A. Ligases | B. Transferases |
| C. Lyases | D. Isomerases |
-
- 80.** This item is based on the following lists of health impacts of amoebiasis and malaria. I. Skin ulcers and hepatosplenomegaly II. Leads to blindness and mental retardation III. Bloody diarrheal fluid due to intestinal cramps IV. Vomiting, convulsion and leads to severe anemia through time Based on the above list, which numbers indicate the health impact of amoebiasis and malaria, respectively? (1)
- | | |
|--------------|---------------|
| A. II and IV | B. I and III |
| C. I and III | D. III and IV |
-
- 81.** During lifting of a heavy object from the ground to certain height, the muscle (1)
- A. contracts due to the flow of Ca^{2+} to sarcoplasmic reticulum which binds actin and myosin.
 - B. relaxes due to the sliding of actin and myosin by utilizing energy from ATP.
 - C. contracts due to the inflow of Ca^{2+} that stimulates binding of actin and myosin.
 - D. relaxes due to the inflow of Ca^{2+} that unbound actin and myosin.
-
- 82.** Which one of the following statements is correct about the role of neurotransmitters? It provides the (1)
- A. synaptic transmission to block its one-way process by stimulating postsynaptic cell.
 - B. endocrine system to produces hormones that are transported in the blood plasma to their target cells.
 - C. nervous system to stimulate their target cells by secreting chemicals directly on to them.
 - D. nervous system to transmit messages between neurons, but not from neurons to muscles.
-
- 83.** Among the following definitions of different ecological levels, which one describes a population? (1)
- A. A group of individuals of the same species living in a separate habitat.
 - B. The major terrestrial characterized by specific vegetation types and physical components.
 - C. A group of interbreeding organisms of the same species living in the same area at the same time.
 - D. The abiotic component of an area without which living things cannot survive.

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- 84.** This item is based on the following description of joints with their role in movement. I. A joint that allows movement in all directions II. A joint that allows movement without rotation III. A joint that allows rotary movement when one bone swivels in a ring IV. A joint that allows limited movement by slipping 2018 Biology First Round V. A joint that allows movement in one direction along one plane only Based on the above information, condyloid joint, pivot joint and hinge joint are respectively represented by numbers (1)
- A. I, II and V. B. II, III and V.
C. II, III and IV. D. I, III and V.
-
- 85.** Consider the following diagram, which indicates the parts of human ear. Identify the labeled part 'I'. (1)
[Figure: A colored anatomical diagram of the human ear showing the outer, middle, and inner ear structures. An arrow labeled with the number 1 points directly to the ear canal, also known as the auditory canal, leading inwards from the pinna towards the eardrum.]
- A. Tympanic membrane (Eardrum) B. Auditory canal
C. Cochlea D. Semicircular canals
-
- 86.** A hemophilic man has two daughters and two sons from neither carrier nor affected woman. One of their sons married to a girl who is heterozygous to hemophilia. What is the probability that one of their daughters become hemophilic? (1)
- A. 50% B. 0%
C. 75% D. 25%
-
- 87.** The scientific names of which one of the following plants is properly written? (1)
- A. Tef - Eragrostis Tef
B. Maize - Zea mays
C. Wheat - triticum aestivum
D. Enset - ensete Ventricosum 2018 Biology First Round
-
- 88.** Which of the following blood vessels carry oxygenated and deoxygenated into and away from the kidneys? (1)
- A. Iliac artery and iliac vein B. Hepatic artery and hepatic vein
C. Renal artery and renal vein D. Pulmonary artery and pulmonary vein
-
- 89.** Grade 10 student placed seeds 'X' and 'Y' in separate containers containing soil and other requirements of germination. From her observation, the cotyledon of seed 'X' remained under the soil during germination, whereas that of seed 'Y' was pushed above the soil surface. What would she conclude about the experiment? (1)
- A. Seed 'X' was most probably monocot whereas seed 'Y' was a dicot.
B. Seed 'Y' has hypogea, whereas seed 'X' has epigeal germination.
C. Differences in the germination was related with availability of water.
D. Light intensity was responsible to grow above and below the soil.
-
- 90.** Which of the following is correct about founder effect in a population? It occurs when (1)
- A. the size of a population is severely reduced due to a sudden accident.
B. new group fully represents genetic diversity of the original population.
C. the new small colony is established from the original population.
D. a natural disaster kills most individuals of the original population.

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- 91.** Which one of the following indigenous practices in Ethiopia is correctly matched with the purpose of plants conservation? (1)
- A. Planting in graveyards - for preserving crop varieties
 - B. Crop rotation - for preserving indigenous woody plants
 - C. Protection in sacred places - preserving genetic resources of vegetables
 - D. Agroforestry - maintaining the genetic diversity of crops and trees
-
- 92.** Which of the following parts of axial skeleton is correctly matched with its function? (1)
- A. Hyoid - protects heart and the lungs
 - B. Cranium - protects the brain
 - C. Sternum - amplify sound wave
 - D. Ribs - holds up human body
-
- 93.** Which property of an enzyme and its description is correctly matched? (1)
- A. Specificity - the ability of an enzyme to catalyze different substrates at a time
 - B. Regulation - controlling activity of an enzyme by activators or inhibitors
 - C. Reversibility - the ability of an enzyme to repair itself after denaturation
 - D. Biocatalytic property - the ability of an enzyme to dissolve in solvents 2018 Biology First Round
-
- 94.** Which of the following best describes the outcome of meiosis in human cells? (1)
- A. Two diploid daughter cells identical to the parent cell
 - B. Four haploid daughter cells, each with half the chromosome number of the parent cell
 - C. Two haploid cells that immediately fuse to form a zygote
 - D. Four diploid cells that undergo further division

— END OF EXAMINATION —